Prazwal Ratti

Web3 & Backend Developer

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About Me

Backend and systems developer working in Rust, Go, TypeScript, and Solidity — building automation tooling, SDKs, and reliability-focused infrastructure. Open-source contributor across Foundry, Convex, Optimism, Cosmos, Compound, and Circle. Solana Turbin3 graduate and Atrium Academy UHI10 graduate. I gravitate to correctness, automation, and code that runs in production; I'd rather ship a working system than a perfect plan.

Open-Source Contributions

Contributed fixes and features to major production codebases in Rust, Go, Solidity, and TypeScript:

- **foundry-rs/foundry (Ethereum tooling, Rust)** — enabled `forge verify-bytecode` to run without a block explorer (#15142).
- **Convex** — hardened the Postgres wire-protocol decoder against malformed messages (#596).
- **Cosmos SDK (Go)** — pruned stale commit-info metadata and made `BaseApp.Close` idempotent (#26561, #26562).
- **aptos-labs/aptos-core (Move VM, Rust)** — validated Move `Identifier` during deserialization for safer module loading (#20056).
- **Circle (arc-node & evm-cctp-contracts)** — hardened node robustness against streaming-overflow DoS, spammer panics, and metrics accuracy (#161); added `ITokenMessenger` interfaces and `minFee` fuzz invariants to the Cross-Chain Transfer Protocol (#108, #109).
- Additional contributions to **Optimism**, **Compound**, **Hyperliquid**, **rust-lang/rust**, **Uniswap v4**, **Noir**, **Oasis**, **Fetch.ai**, **Stellar** & **Keplr**.

Projects

agentctl | TypeScript, Solidity, Rust 2026

- Policy & observability layer for autonomous AI-agent payments: sits between an agent and its wallet, enforcing spending policy and producing a tamper-evident audit trail for every payment.
- Three-layer decision engine (deterministic policy + EWMA/Isolation-Forest anomaly detection + LLM intent reconciliation $\rightarrow$ allow/escalate/block); Merkle-batched audit log anchored on-chain; OpenTelemetry tracing.

Assay | Solidity, TypeScript 2026

- Uniswap v4 hook that prices each swap by its adverse-selection risk instead of a flat pool-wide fee; derived from LVR ($\sigma^2 V/8$) research to protect LPs from informed-flow bleed.
- **Uniswap Foundation Prize Winner** — Atrium Academy UHI10 hookathon. Live demo: UHI10 Demo Day.

Apiforge | Rust 2026

- Production-grade CLI automating the full API release pipeline (version bump $\rightarrow$ Docker $\rightarrow$ Kubernetes $\rightarrow$ health checks) with automatic rollback, dry-run, and audit logging.
- Integrates AWS ECR, Docker Hub, GitHub Releases, Kubernetes & Slack; published on crates.io — **200+ downloads**.

SwapKit | TypeScript, Rust, Solidity 2026

- Intent-based liquidity SDK exposing one `swap(intent)` call that abstracts 1inch Fusion+, Uniswap v4 hooks, and Paraswap; MEV-risk simulation and cross-chain routing; **220+ tests**; **3,500+ npm downloads**.

Education

Vellore Institute of Technology (VIT), Chennai | B.Tech, Computer Science (Core) — CGPA 8.10 / 10 2023 – 2027

FIITJEE | Class XII — 93% • Class X — 99%

Skills

Languages: Rust, Go, Solidity, TypeScript, JavaScript, Python, C, C++, Noir, Circom, SQL, Bash

Backend: Node.js, Express, Flask, REST, gRPC, GraphQL, WebSockets, PostgreSQL, MongoDB, Redis, Kafka, Microservices

Blockchain / Web3: Ethereum, Solana, Foundry, Ethers.js, Wagmi, Chainlink, The Graph, OpenZeppelin, IPFS, zkSync, Arbitrum, Optimism, Sui, ZK Proofs

DevOps & Tools: Docker, Kubernetes, AWS (ECR), GitHub Actions, CI/CD, Linux, Git, OpenTelemetry